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Lab report **1**

1. Purpose of the Lab

The aim of this practical session was to prepare a working C++ development environment on a Windows operating system using Visual Studio Code. The setup process involved installing the required compiler, adjusting system settings, and adding supportive tools to enable smooth program development and execution.

2. Required Software and Tools

The following applications and components were used:

- Visual Studio Code as the code editor
- MinGW-w64 as the C++ compiler
- C/C++ Extension provided by Microsoft
- Code Runner Extension developed by Jun Han

3. Installation and Configuration Steps

Step 1: Installing Visual Studio Code

- The installer was downloaded from the official website:
<https://code.visualstudio.com/Download>.
 - The setup file was executed, and installation was completed by following the instructions shown on the screen.
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Step 2: Installing the Compiler

- The MinGW-w64 package was obtained from SourceForge.
 - The appropriate version was selected and installed on the system.
 - Necessary installation options were confirmed to complete the setup successfully.
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Step 3: Setting Up System Path

- The directory containing the compiler was identified.
 - This location was added to the system's Path variable inside the environment settings.
 - The configuration was saved so the compiler could be accessed from any directory in the command line and editor.
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Step 4: Adding Extensions in VS Code

- The C/C++ extension by Microsoft was installed from the extensions marketplace.
 - The Code Runner extension was also installed to simplify program execution.
 - Visual Studio Code was restarted to ensure the extensions worked properly.
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4. Practical Assignment

The assignment included the following tasks:

1. Creating a team of three participants.
2. Successfully completing the installation and configuration process.
3. Writing and executing a basic C++ program that prints the names of group members.

Example Program

```
#include <iostream>
using namespace std;

int main() {
    cout << "Team Members:" << endl;
    cout << "1. Naeem Ulhaq" << endl;
    cout << "2. [Second Member]" << endl;
    cout << "3. [Third Member]" << endl;
    return 0;
}
```

5. Program Output

```
Team Members:
1. Naeem Ulhaq
2. [Second Member]
3. [Third Member]
```

6. Findings

- The installation procedure was simple but required accurate configuration of system settings.
 - If the compiler path was not added to the environment variables, C++ programs could not be compiled successfully.
 - The Code Runner extension made program execution faster by allowing direct running inside the editor.
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7. Final Remarks

The objective of this laboratory activity was successfully accomplished. The C++ development environment was properly installed and configured in Visual Studio Code, and a sample program was executed without errors. This setup will support future programming exercises and academic projects.